higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 2

22 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1	1.1.1	Structural wall		
	1.1.2	Racking back		
	1.1.3	An anchor		
	1.1.4	Concrete		
	1.1.5	Mortar		
			(5 × 1)	(5)
1.2	1.2.1	True		
	1.2.2	False		
	1.2.3	False		
	1.2.4	True		
	1.2.5	True		
			(5 × 1)	(5)
1.3	1.3.1	A		
	1.3.2	B		
	1.3.3	C		
	1.3.4	B		
	1.3.5	C		
			(5 × 1)	(5)
				[15]

QUESTION 2: INTRODUCTION AND BACKGROUND

2.1 A block is a concrete masonry unit✓ which has a length of between 300 mm and 650 mm, a width of between 90 mm and 300 mm, and a height of between 100 mm and 300 mm.✓
Bricks usually refer to fired clay✓ modules of masonry units. A brick is usually any masonry unit which is not a block, and is usually sized as 76 × 110 × 220.✓ (2 × 2) (4)

2.2 A masonry unit is a unit that is intended for the construction of bonded-masonry walling. (1)

2.3 73 mm (1)
[6]

QUESTION 3: SETTING OUT MASONRY WALLS

3.1
- 3:4:5 method
- Diagonal method
- Usage of a theodolite
- Usage of a steel square (4)

3.2 3.2.1
- Hand hawk
- Builder's square
- Spirit level
- Straight edge
- Block brush
- Inside-corner tool (Any 4 × 1) (4)

3.2.2
- Four-pound hammer
- Steel pegs
- Straight edge
- Builder's square (4)

3.2.3
- Joint ruler
- Racking tool (2)

3.3 3.3.1 No,✓ these walls are non-structural walls✓ and do not bear the weight of the roof or any other part of the house.✓ They simply support their own weight.✓ (4)

3.3.2
- Length = 7 meters
- Width = 6 meters (2)

3.3.3 The thickness of the walls✓ are missing as well as the height of the superstructure. ✓ (2)

[22]

QUESTION 4: BASIC CONSTRUCTION OF MASONRY WALLS.

4.1 4.1.1 In cavity walls, DPC is laid across the full width of the wall, including the cavity,✓ and is stepped down one course in the cavity over a mortar fillet.✓ DPC shall be used above all window and door openings in cavity walls✓ (Any 2 × 1)

4.1.2 DPC is not used in free-standing walls✓ because the walls are self-supporting.✓

4.1.3 Must be inserted or laid in accordance with the drawings.✓ Apply mortar bed on top and bottom of the DPC and follow the shape of the building. It is usually laid above ground level.✓ (Any 3 × 2) (6)

4.2 DPC forms a barrier to water.✓ It prevents the upwards or sideways movement of water through a wall.✓ (2)

4.3 **A single-leaf wall** consists of a single row of bricks laid end to end (in stretcher pattern).✓ A single-leaf wall in brickwork is usually 110 mm thick.✓

A cavity wall is similar to a double-leaf wall, except that there is a hollow space between the leaves.✓ The two leaves in a cavity wall are tied together by means of wall ties, and a cavity wall has weep holes.✓ (2 × 2) (4)

4.4 Bed joint reinforcing (1)

4.5 Brickforce makes masonry walls stronger✓ and they are less likely to crack because of natural movements of the wall✓ or normal movements caused by either temperature changes or the natural expansion or contraction of brick or concrete products. It controls movement and keeps walls stable.✓ It is also a valuable aid in reinforcing parts of the wall where windows or doors have been put in.✓ (4)

4.6 A – Concave joint
B – Vee joint
C – Weathered joint
D – Flush joint
E – Raked joint
F – Struck joint (6)

4.7
- 190 × 90 × 90
- 290 × 90 × 90
- 390 × 90 × 190
- 390 × 190 × 190 (Any 3 × 1) (3)

4.8 Weepholes in walls are designed to drain away moisture. (1)

4.9 Butterfly wall ties. (1)
[28]

QUESTION 5: ADVANCED CONSTRUCTION OF MASONRY WALLS

5.1

5.1.1	G
5.1.2	I
5.1.3	A
5.1.4	C
5.1.5	B
5.1.6	H
5.1.7	D
5.1.8	K
5.1.9	J
5.1.10	L

(10 × 1) (10)

5.2 Create a timber frame on the inside of the opening for the lintel to rest on.✓ Lay mortar over the frame of the door/window and ensure that the wall plate is level with the top of the lintel. Use a spirit level to ensure that the lintel is level.✓ Place DPC over the concrete lintel and put sealant under the lintel to prevent moisture entering.✓ Determine the minimum number of courses.✓ Lay each course with brickforce in between the courses.✓ Ensure that each course is level and plumb.✓
The cavity (if a cavity wall) must be filled with mortar or infill concrete.✓
Wait seven days (or more) and remove the timber frame.✓ (8)

5.3
- The slope/pitch of the roof.
- The type of material the roof is made of.
- The minimum truss spacing. (3)

5.4
- The ties join two walls so that any movement in one is transferred to the other.
- Prevents cracking and structural damage. (2)

5.5 DPC must be placed on the course below the window frame.✓ It must project 200 mm on either side of the window frame.✓ Place sealant under the DPC.✓ (3)

5.6 Vertical control joints in non-reinforced masonry must generally be positioned where a concentration of or changes in stress may occur. For example, at openings,✓ major changes in wall height, and changes in wall thickness.✓ Control joints can also be located between openings.✓ (3)

[29]

TOTAL: 100

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